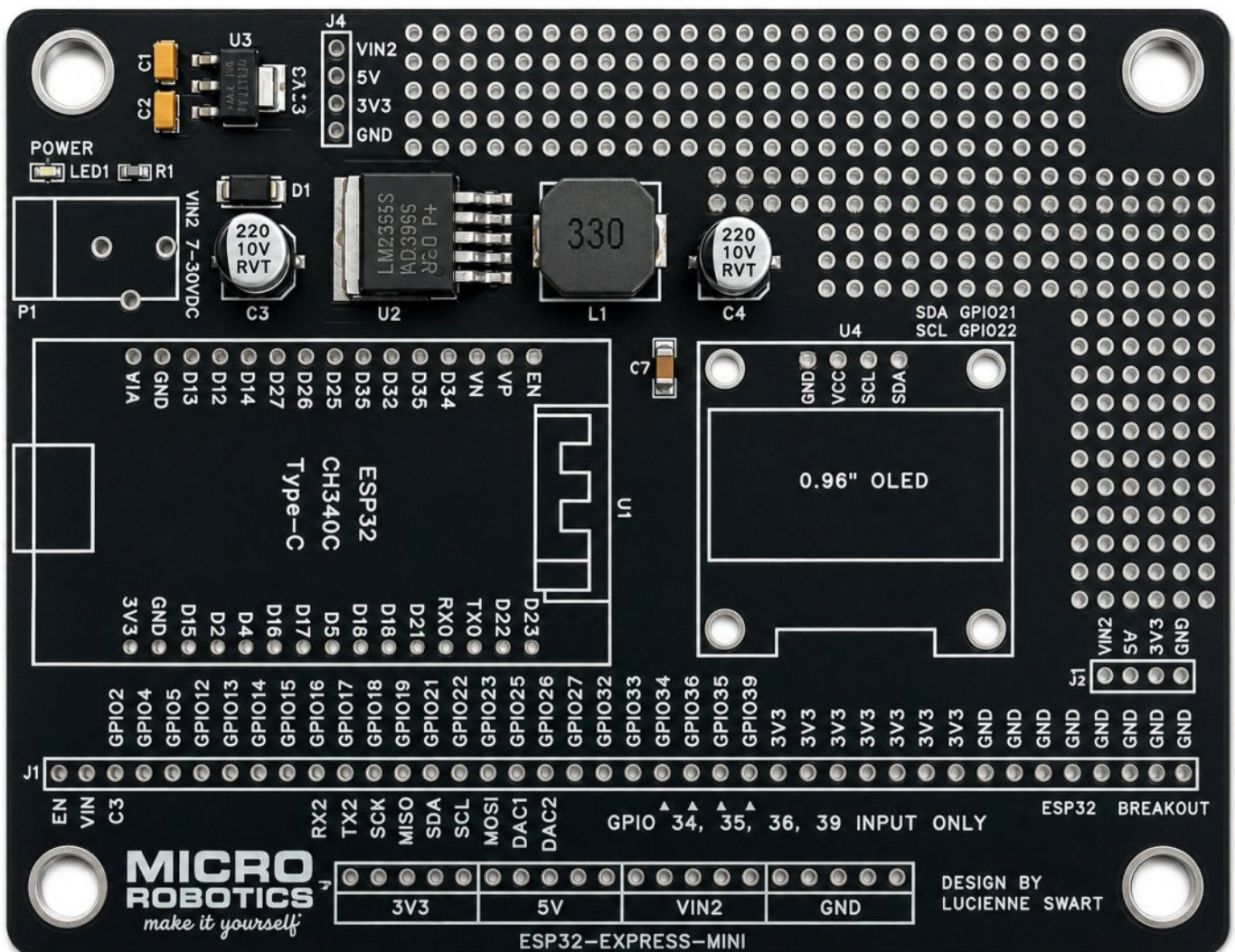


ESP32 IoT Controller PC Board

COMPONENT
IDENTIFICATION

1. ESP32 Main Board

The ESP32 PCB houses all the main components, including the ESP32 microcontroller module, voltage regulators, and the OLED display.



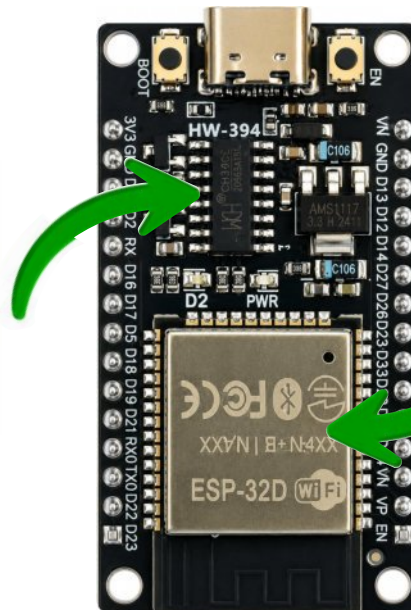
2. ESP32 Microcontroller Module (30 Pins)



The 30-pin ESP32 development board is the main processing unit of the kit, providing Wi-Fi and Bluetooth connectivity for IoT applications. It features a CH340 USB-to-Serial interface for programming and debugging from a computer, while its versatile GPIO pins enable easy connection to sensors, displays, actuators and other electronic peripherals.



The module use a
CH340 USB Chipset



ESP32
Microcontroller

Why is this important

Knowing the USB-to-Serial chipset ensures you install the correct driver for reliable firmware uploads, serial communication, automatic programming, and compatibility with Windows, macOS, and Linux.

3. OLED Display Screen



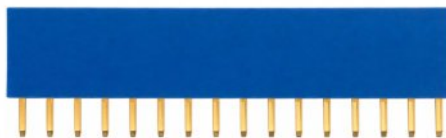
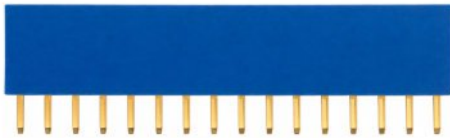
Display Type: OLED
Display Driver: SSD1315
Display Size: 0.96"
Resolution: 128 × 64 pixels
Communication Interface: I²C

Take Note: Use the correct SSD1315 display driver when writing software for this display.

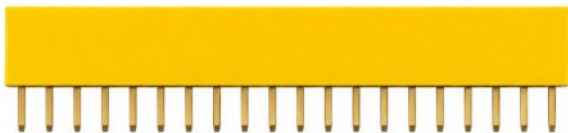
4. Connectors



1 x DC JACK Connector



2 x Female Header 15 Pin Blue



1 x Female Header 20 Pin Black



1 x Female Header 40 Pin Black

5. AC Adapter - Output 9V 2A

USB A to C cable - programming ESP32



NEXT
ASSEMBLY OF BOARD

